

Journal Supplementary B — Full related-work taxonomy

Companion to: *Beyond Accuracy* (Section II)

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A.1 Scope

This supplement expands the related-work review of journal Section II with thematic clustering and direct method-level comparison. We organise the literature into four bands: (i) foundational topic modelling, (ii) neural topic models, (iii) HSI classification and unmixing including topic-flavoured work, and (iv) the evaluation methodology literature this paper draws on.

B.2 Foundational topic modelling

LDA and its inference variants. Blei, Ng, and Jordan [1] introduced the canonical Dirichlet prior-over-topic-mixtures generative model with variational EM for inference. Griffiths and Steyvers [2] developed the collapsed Gibbs alternative which avoids the variational distribution but has higher per-iteration cost. Hoffman, Blei, and Bach [3] introduced online variational Bayes with natural-gradient ascent on the global Dirichlet pseudo-counts, enabling LDA fits on corpora that do not fit in memory; the scikit-learn implementation [4] we use in this paper follows the Hoffman 2010 recipe.

Non-parametric extensions. Teh, Jordan, Beal, and Blei [5] introduced the Hierarchical Dirichlet Process as a non-parametric prior that lets the data choose the topic count. We do not use HDP here because the band-frequency tokenisation produces relatively short documents on which the HDP’s truncation-level estimate is unstable, and the experimental budget would have multiplied without proportional information gain.

Topic interpretability tooling. Sievert and Shirley [6] introduced LDAvis, the canonical interactive interpretation tool for topic-word distributions, with the relevance score $\lambda \phi_{k,b} + (1 - \lambda) \log \phi_{k,b}/p_b$. We use $\lambda = 0.5$ for the per-topic top-word extraction. Taddy [7] provided the formal relationship between topic-word and word-topic probabilities used in the journal’s F-7 axis.

C.3 Neural topic models

ProdLDA. Srivastava and Sutton [8] introduced ProdLDA by reformulating LDA in a product-of-experts form and combining it with the auto-encoding variational Bayes framework of Kingma and Welling [9]. The Laplace approximation to the Dirichlet prior is the technical core that makes amortised inference tractable.

ETM. Dieng, Ruiz, and Blei [10] introduced the Embedded Topic Model, which decomposes the topic-word distribution into an $V \times E$ word-embedding matrix and a $K \times E$ topic-embedding matrix and softmax-normalises their inner product. The low-rank decomposition lets rare words inherit the topic associations of their semantic neighbours.

Neural variational topic models. Miao, Yu, and Blunsom [11] introduced the broader family of neural variational document models (NVDM) of which ProdLDA can be seen as a constrained variant; Higgins et al. [12] provided the β -VAE perspective that the deep-baseline ladder in the companion code repository draws on.

What is missing. The published neural-topic-model literature for HSI is thin. ProdLDA and ETM have been applied to text corpora extensively but to HSI rarely; the small body of HSI papers that cite either method typically applies them on a single scene with a single seed. The seed-stability swarms we report (axis F-3, journal Figure 4) are, to our knowledge, the first multi-seed ProdLDA and ETM evaluations on the six labelled benchmarks.

D.4 HSI classification, unmixing, and topic-flavoured work

Linear unmixing. Nascimento and Bioucas-Dias [13] introduced the VCA endmember extraction algorithm. The broader unmixing literature is reviewed by Bioucas-Dias et al. [14]. Linear unmixing shares with LDA the non-negative basis structure but differs in two operative ways: (a) unmixing assumes a fixed endmember count determined by physical reasoning, LDA admits K as a free hyperparameter; (b) unmixing produces a per-pixel abundance vector that is constrained to sum to one across endmembers, LDA produces a per-document Dirichlet posterior whose mean satisfies the same sum constraint asymptotically but admits a richer family of finite-data shapes.

Deep spectral-spatial classification. Plaza et al. [15] reviewed the first decade of HSI classification, Camps-Valls et al. [16] the following half-decade with explicit attention to manifold and kernel methods. Chen et al. [17] introduced the first systematic deep architectures for HSI classification and Paoletti et al. [18] surveyed the subsequent five years. The OA-as-summary discipline in this literature is well-developed and our journal paper does not challenge it; the challenge is that topic-model adaptations inherited the discipline without inheriting the inductive bias that makes it well-suited to the supervised classification task.

Topic models on HSI. Zou and Zare [19] introduced partial-membership LDA on a small set of HSI benchmarks. Liu et al. [20] explored a spectral-derivative tokenisation. Both papers focus on improving the classification OA of the topic representation; neither reports the cross-axis diagnostics that the journal paper’s twelve-axis framework recommends.

E.5 Evaluation methodology

Topic-model coherence. Röder, Both, and Hinneburg [21] surveyed coherence measures and showed that the sliding-window c_v measure tracks human judgements of interpretability more reliably than perplexity. We use c_v , c_{NPMI} , and U-Mass for axis F-2.

Clustering comparison. Hubert and Arabie [22] formalised the adjusted Rand index. Vinh, Epps, and Bailey [23] extended the ARI toolbox with the AMI and normalised information variants and discussed when each is preferable.

Bipartite matching. Kuhn [24] introduced the Hungarian algorithm for the maximum-weight bipartite matching problem that the topic-id permutation step (axis F-8) solves. The numerical implementation we use is in SciPy [25, 26].

Reproducibility discipline. Pineau et al. [27] argued for ML reproducibility checklists. The journal paper’s commitment to a single canonical fit + per-axis deterministic builder + JSON artefact + smoke test is one instantiation of that checklist.

Information-theoretic complements. Koltchinskii [28] provides the empirical-process foundation for the stability bounds used in seed-stability analysis. Achanta et al. [29] introduced SLIC, which we use as a cross-method-agreement baseline in axis F-6; Felzenszwalb and Huttenlocher [30] provided the spatial-graph segmentation we use as the non-spectral counterpart.

F.6 Where this paper sits in the taxonomy

The journal paper does not introduce a new topic model, a new tokenisation, or a new inference procedure. It contributes a *multi-axis evaluation framework* that subsumes the above-cited evaluation tools (Röder coherence, Hubert–Arabie ARI, Kuhn matching, scikit-learn LDA, SciPy hypothesis tests) into a single artefact pipeline that yields a per-axis numeric per benchmark scene plus a hierarchical-Bayesian summary across scenes. The contribution is methodological rather than algorithmic.

References

- [1] D. M. Blei, A. Y. Ng, and M. I. Jordan, “Latent Dirichlet allocation,” *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003. [Online]. Available: <https://jmlr.org/papers/v3/blei03a.html>
- [2] T. L. Griffiths and M. Steyvers, “Finding scientific topics,” *Proceedings of the National Academy of Sciences*, vol. 101, no. Suppl. 1, pp. 5228–5235, 2004.
- [3] M. D. Hoffman, D. M. Blei, and F. R. Bach, “Online learning for latent Dirichlet allocation,” vol. 23, pp. 856–864, 2010. [Online]. Available: <https://papers.nips.cc/paper/3902-online-learning-for-latent-dirichlet-allocation>
- [4] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
- [5] Y. W. Teh, M. I. Jordan, M. J. Beal, and D. M. Blei, “Hierarchical Dirichlet processes,” *Journal of the American Statistical Association*, vol. 101, no. 476, pp. 1566–1581, 2006.
- [6] C. Sievert and K. E. Shirley, “LDAvis: A method for visualizing and interpreting topics,” in *Proceedings of the Workshop on Interactive Language Learning, Visualization, and Interfaces*, 2014, pp. 63–70.
- [7] M. Taddy, “On estimation and selection for topic models,” in *Proceedings of the Fifteenth International Conference on Artificial Intelligence and Statistics*, 2012, pp. 1184–1193. [Online]. Available: <http://proceedings.mlr.press/v22/taddy12.html>
- [8] A. Srivastava and C. Sutton, “Autoencoding variational inference for topic models,” in *International Conference on Learning Representations (ICLR)*, 2017, introduces ProdLDA. arXiv:1703.01488. [Online]. Available: <https://openreview.net/forum?id=BybtVK9lg>
- [9] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *International Conference on Learning Representations (ICLR)*, 2014. [Online]. Available: <https://arxiv.org/abs/1312.6114>
- [10] A. B. Dieng, F. J. R. Ruiz, and D. M. Blei, “Topic modeling in embedding spaces,” *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 439–453, 2020.

- [11] Y. Miao, L. Yu, and P. Blunsom, “Neural variational inference for text processing,” in *International Conference on Machine Learning (ICML)*, 2016, pp. 1727–1736. [Online]. Available: <http://proceedings.mlr.press/v48/miao16.html>
- [12] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, “beta-VAE: Learning basic visual concepts with a constrained variational framework,” in *International Conference on Learning Representations (ICLR)*, 2017. [Online]. Available: <https://openreview.net/forum?id=Sy2fzU9gl>
- [13] J. M. P. Nascimento and J. M. Bioucas-Dias, “Vertex component analysis: A fast algorithm to unmix hyperspectral data,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 43, no. 4, pp. 898–910, 2005.
- [14] J. M. Bioucas-Dias, A. Plaza, N. Dobigeon, M. Parente, Q. Du, P. Gader, and J. Chanussot, “Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 5, no. 2, pp. 354–379, 2012.
- [15] A. Plaza, J. A. Benediktsson, J. W. Boardman, J. Brazile, L. Bruzzone, G. Camps-Valls, J. Chanussot, M. Fauvel, P. Gamba, A. Gualtieri, M. Marconcini, J. C. Tilton, and G. Trianni, “Recent advances in techniques for hyperspectral image processing,” *Remote Sensing of Environment*, vol. 113, no. Supplement 1, pp. S110–S122, 2009.
- [16] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, “Advances in hyperspectral image classification: Earth monitoring with statistical learning methods,” *IEEE Signal Processing Magazine*, vol. 31, no. 1, pp. 45–54, 2014.
- [17] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, “Deep learning-based classification of hyperspectral data,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 7, no. 6, pp. 2094–2107, 2014.
- [18] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, “Deep learning classifiers for hyperspectral imaging: A review,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 158, pp. 279–317, 2019.
- [19] S. Zou and A. Zare, “A continuous-space partial-membership latent Dirichlet allocation approach for subpixel hyperspectral imagery,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 10, pp. 5840–5853, 2017.
- [20] L. Liu, C. Wang, B. Du, and H. Lai, “A latent Dirichlet allocation-based local spectral-spatial hyperspectral image classification method,” *IEEE Geoscience and Remote Sensing Letters*, vol. 16, no. 6, pp. 935–939, 2019.
- [21] M. Röder, A. Both, and A. Hinneburg, “Exploring the space of topic coherence measures,” in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining (WSDM)*, 2015, pp. 399–408.
- [22] L. Hubert and P. Arabie, “Comparing partitions,” *Journal of Classification*, vol. 2, no. 1, pp. 193–218, 1985.
- [23] N. X. Vinh, J. Epps, and J. Bailey, “Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance,” *Journal of Machine Learning Research*, vol. 11, pp. 2837–2854, 2010. [Online]. Available: <https://jmlr.org/papers/v11/vinh10a.html>
- [24] H. W. Kuhn, “The Hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

- [25] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, and P. van Mulbregt, “SciPy 1.0: Fundamental algorithms for scientific computing in Python,” *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.
- [26] SciPy Developers, “`scipy.optimize.linear_sum_assignment` — SciPy reference,” Online; SciPy v1.13 reference manual, 2024. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html
- [27] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d’Alché Buc, E. Fox, and H. Larochelle, “Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program),” *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: <https://jmlr.org/papers/v22/20-303.html>
- [28] V. Koltchinskii and D. Panchenko, “Empirical margin distributions and bounding the generalization error of combined classifiers,” *The Annals of Statistics*, vol. 30, no. 1, pp. 1–50, 2002.
- [29] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, “SLIC superpixels compared to state-of-the-art superpixel methods,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, no. 11, pp. 2274–2282, 2012.
- [30] P. F. Felzenszwalb and D. P. Huttenlocher, “Efficient graph-based image segmentation,” *International Journal of Computer Vision*, vol. 59, no. 2, pp. 167–181, 2004.